

Supplementary physiological benchmarking and comparative genomic context

Supplementary Figure S2

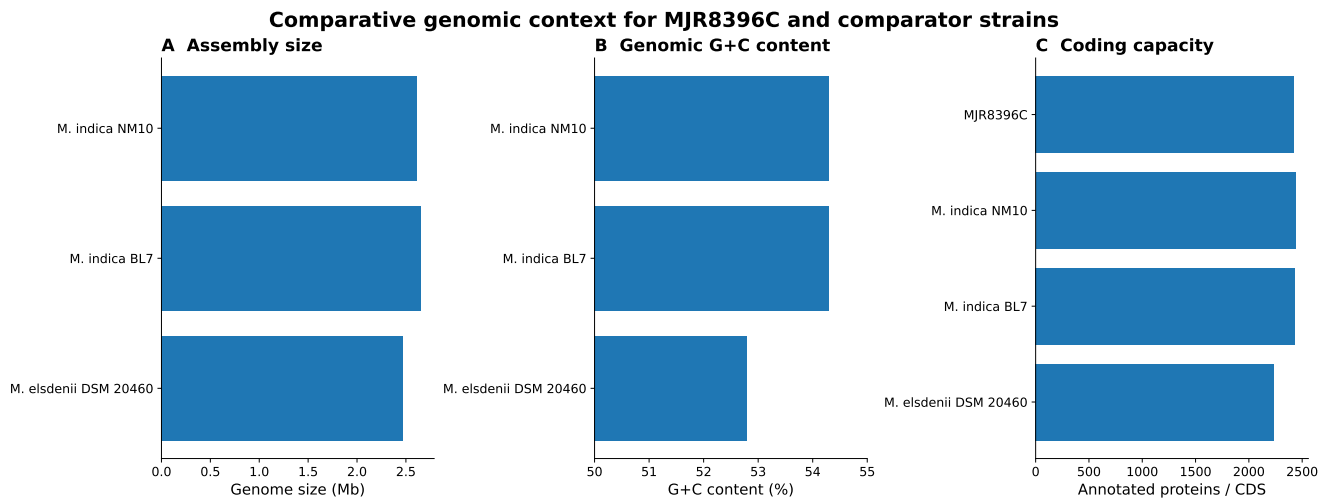


Figure 1: **Comparative genomic context for MJR8396C and physiological comparator strains.** Assembly size and G+C content are shown where uniformly reported in the source literature; coding capacity is shown as annotated proteins or coding sequences and should therefore be interpreted as an approximate contextual comparison rather than a uniformly reannotated pangenome analysis.

Supplementary Table S11. Benchmark performance metrics

Comparison	Spearman ρ	Pearson r	RMSE (h^{-1})
MJR8396C GEM versus experimental growth	0.50	0.91	0.219
Published iME375 versus experimental growth	1.00	0.99	0.012

Supplementary Table S14. Comparative genomic context

Strain	Habitat	Assembly	Genome (Mb)	G+C (%)	Proteins/CDS	Role
MJR8396C (MJR9396C in genome report)	Human vagina	GCA_001546855.1 / LRVC00000000	–	–	2422	Target strain
<i>M. indica</i> NM10	Human feces	APHY00000000	2.615	54.3	2436	Human-gut comparator
<i>M. indica</i> BL7	Human feces	APHX00000000	2.656	54.3	2432	Human-gut comparator
<i>M. elsdenii</i> DSM 20460	Rumen	HE576794	2.472	52.8	2236	Physiological comparator

Interpretive scope. NM10 and BL7 were compared directly in the original human-gut isolate study and were reported to be highly similar to one another but substantially different from the ruminal DSM 20460 proteome. MJR8396C is a vaginal HMP reference isolate and was not included in that

historical comparative analysis. Accordingly, these data establish ecological and genomic context but do not constitute formal ANI, dDDH, or pangenome evidence for MJR8396C relatedness.